

Ishan Maitra

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PROFESSIONAL SUMMARY

AI Developer and B.Tech graduate in Artificial Intelligence & Machine Learning with hands-on production experience across LLM orchestration, autonomous agent pipelines, and Python-based numerical validation. Demonstrated expertise in NumPy, Pandas, and SciPy for data-driven simulations, with a proven record of improving AI reasoning alignment by **25%** through structured data labeling, high-integrity scoring frameworks, and rigorous testing protocols. Builder of multiple production-grade AI systems — including gesture-controlled OS interfaces, voice assistants, and Android AI applications — distributed under the Maitra Brand ecosystem.

TECHNICAL SKILLS

Languages & Paradigms: Python (Expert), JavaScript (ES6+), Java, Kotlin, SQL, C/C++, Software Engineering Principles, Clean Architecture

AI / ML Libraries: NumPy, Pandas, SciPy, Scikit-learn, MediaPipe, OpenCV, TensorFlow (basics), Data Annotation & Labeling Toolchains

AI Specialization: LLM Orchestration, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, RLHF, Data Labeling, AI Alignment

Mobile & Frontend: Jetpack Compose, Android (MVVM, Clean Arch, Hilt, Room), React.js, HTML/CSS

Cloud & DevOps: Google Cloud Platform (GCP), Git/GitHub, Linux, Docker (basics), CI/CD Pipelines, Agile Methodologies

Embedded Systems: Arduino, ESP32, IoT Integration, Motor Controllers, Sensor Fusion

Languages (Natural): English — Fluent/C1, Bengali — Native, Hindi — Professional Working Proficiency

PROFESSIONAL EXPERIENCE

Cloud AI Developer Google Cloud Projects

Jan 2025 – Present

- Designed and deployed autonomous multi-agent workflows on Google Cloud Platform, achieving a **20% improvement** in end-to-end task execution efficiency through parallelised agent orchestration and structured reasoning chains.
- Evaluated over **100 AI simulations** for scientific integrity and logical correctness, enforcing alignment with industry-standard validation protocols and multi-step reasoning benchmarks.
- Engineered a suite of Python-based automated **Software Engineering tools** — leveraging NumPy and SciPy — that reduced simulation testing cycle time by **30%** and decreased manual validation overhead across repeated experimental runs.
- Contributed to LLM prompt-alignment research by designing structured scoring rubrics that measurably improved AI output quality and factual consistency across adversarial test cases.

Founder & Lead AI/Software Developer Maitra Web Tech & WebX

Jan 2024 – Present

- Architected and shipped **MaitraGPT**, a production Android AI assistant (Jetpack Compose, Hilt, Room, Gemini API, TTS) with a cyberpunk UI — managing the full engineering lifecycle from design through CI/CD deployment.
- Established high-integrity **data labeling** and annotation workflows for LLM fine-tuning, achieving a **98% structured-output accuracy rate** across all annotated datasets.
- Built **ARIA**, a Hinglish-native Windows voice assistant (Python, Gemini NLU, Tkinter) with deep personality configuration, contextual memory, and real-time speech synthesis, reducing repetitive desktop task overhead by an estimated 40%.
- Developed the **Maitra Neural Control System (MNCS)**, a touchless computer-interaction platform using MediaPipe and PyAutoGUI, enabling full OS control via hand-gesture recognition with sub-50 ms latency.
- Validated **50+ complex AI solutions** for logical correctness and scientific precision using Python (NumPy, Pandas), maintaining documentation of validation outcomes for reproducibility.
- Delivered client-facing full-stack web projects including an AI-powered resume builder, a fake-news detection widget, and a premium industrial company website featuring 3D/animated aesthetics — each within agreed delivery timelines.

KEY PROJECTS

Aero — Production Android Weather App *Kotlin, Jetpack Compose, Clean Architecture, OpenWeatherMap API* 2025

- Delivered a fully production-ready Android weather application using Clean Architecture (Data / Domain / Presentation layers), Kotlin coroutines, and Jetpack Compose UI — resolving Gradle wrapper compatibility issues to achieve successful physical-device deployment.

Maitra Neural Control System (MNCS) *Python, MediaPipe, OpenCV, PyAutoGUI, Kalman Filter* 2024

- Designed a physics-informed OS control system that uses hand-landmark tracking (MediaPipe) combined with Kalman filtering for real-time gesture smoothing, enabling zero-touch interaction with Windows 11 at sub-50 ms end-to-end latency.

High-Integrity AI Alignment Engine *Python, NumPy, Pandas, Structured Scoring Frameworks* 2024

- Built a structured data-labeling and reasoning-validation platform that applies multi-criteria scoring rubrics to evaluate LLM output correctness for complex logical and scientific queries, informing RLHF improvement cycles.

MaitraAI v2 — Self-Hosted Personal AI Ecosystem *Python, ChromaDB, Mistral 7B, Whisper, Gemini, RAG* 2023–2024

- Engineered a self-hosted AI assistant with RAG over personal documents (ChromaDB + Mistral 7B), OS-level control, multi-platform messaging bridges (WhatsApp, Telegram, Discord, Slack), and Hinglish NLU — a ground-up local-first AI infrastructure stack.

note.ai — AI Meeting Note Taker *React, Claude API, REST* 2024

- Built a real-time meeting note-taking web application powered by the Anthropic Claude API, featuring automatic structuring of action items, summaries, and decisions from raw transcript input.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning 2021 – 2025
Budge Budge Institute of Technology (BBIT), Kolkata, West Bengal

CERTIFICATIONS & CREDENTIALS

Google Cloud — AI & Machine Learning Practitioner Path 2025
GeeksforGeeks 60-Day Problem of the Day Challenge (Java, Python, C++) 2024
Competitive Programming — Active on LeetCode (*sequential problem-solving, DSA notebook*) Ongoing

ACHIEVEMENTS & RECOGNITION

- Improved AI reasoning-alignment metrics by **25%** through systematic data labeling, structured scoring, and adversarial validation — documented across 100+ simulation evaluations.
- Grew professional audience on **LinkedIn & X (@maitra_ishan)** by consistently publishing technical content about personal AI projects, developer tools, and ML system design.
- Completed the **GeeksforGeeks 60-Day POTD Challenge**, demonstrating consistent algorithmic problem-solving discipline in Java, Python, and C++ simultaneously.
- Shipped **6+ production-grade applications** spanning Android, Windows, and Web platforms within a single year under the Maitra Brand, from conception to public GitHub release.